RAAGA SINDHU

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PROFESSIONAL SUMMARY

Data Scientist with 4+ years developing machine learning models, ETL/ELT pipelines, and data quality frameworks across large-scale enterprise datasets. Experienced in statistical modeling, NLP, predictive analytics, and model deployment — with a track record of translating complex analytical findings into measurable business outcomes. Proven ability to collaborate with cross-functional teams and senior stakeholders to design and deliver production-ready AI/ML solutions.

CORE COMPETENCIES

Machine Learning & Predictive Modeling • Statistical & Econometric Modeling • NLP & Sentiment Analysis • Feature Engineering • Hypothesis Testing & Experimentation • ETL/ELT Pipeline Development • MLOps & Model Deployment • A/B Testing & Model Validation • Data Quality & Anomaly Detection • Model Monitoring & Drift Detection • Python • SQL • Vertex AI • BigQuery • Snowflake • Deep Learning • Data Wrangling • Cross-functional Collaboration • Stakeholder Communication • Agile Delivery

EXPERIENCE

Western Governors University – Salt Lake City, UT **Mar 2025 – Jan 2026**

Senior Research Analyst

- Directed enterprise-scale analytics and ML initiatives across 10M+ labor-market records using Python, SQL, and Snowflake - building data pipelines and predictive models to inform institutional strategy, program design, and investment decisions
- Architected a skill intelligence system using regression and clustering to analyze 23,000+ occupational skills, enabling leadership to identify high-demand capabilities 5X faster
- Developed econometric and statistical forecasting models with advanced feature engineering, achieving 90%+ accuracy in market demand estimation; rigorously validated assumptions and tested for sampling biases and data coverage gaps before deployment
- Built AI-driven NLP pipelines for sentiment analysis and topic modeling on 20K+ feedback records, converting qualitative data into metrics that improved decision speed by 25%
- Integrated and reconciled disparate external data sources (Lightcast, BLS, NCES) with institutional data, enforcing quality standards and lineage tracking - boosting analytics accuracy by 20% and reducing data preparation time by 40%

Northeastern University – Boston, MA **Jan 2024 - Aug 2024**

Machine Learning Analyst – NASA

- Constructed an end-to-end ML system for well-location prediction using geospatial and operational data, applying classification models and neural network architectures to achieve 87% accuracy; validated models against alternative approaches and business-defined KPIs
- Engineered distributed ETL/ELT pipelines using Dask and Parquet for large-scale data ingestion, reducing processing time from 45 minutes to seconds; implemented MLOps workflows for model monitoring and CI/CD-aligned deployment on Vertex AI with outputs in BigQuery
- Conducted A/B testing and model validation to measure performance gains, raising reliability by 20%
- Designed interactive Tableau dashboards translating geospatial model outputs into clear, actionable insights for stakeholders

Northeastern University – Boston, MA **Sep 2023 – Dec 2023**

Graduate Teaching Assistant – Data Visualization

- Led workshops on data wrangling, analysis, and visualization, improving student project quality by 15%
- Mentored 100+ students on Tableau, Power BI, and data storytelling workflows

Tata Consultancy Services – Nagpur, India **Jul 2021 - Aug 2022**

Data Analyst – Business Intelligence

- Resolved 95% of Salesforce data issues within SLA, ensuring reliable downstream reporting for leadership teams
- Automated Salesforce-JIRA workflows, reducing manual effort by 30% and strengthening demand planning for procurement
- Analyzed large operational datasets using SQL and Excel; created KPI dashboards that increased forecast accuracy by 25%

EDUCATION

Northeastern University - Boston, MA **Dec 2024**
Master of Science in Data Analytics Engineering

Sreenidhi Institute of Science and Technology - Hyderabad, India **May 2021**
Bachelor of Science in Electronics and Computer Engineering

TECHNICAL SKILLS

Programming & Libraries: Python (Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn, Plotly) • SQL • R
Machine Learning & AI: Predictive Modeling • Regression & Classification • Model Evaluation & Validation • NLP & Sentiment Analysis • A/B Testing • MLOps
Data Engineering: ETL/ELT Pipelines • Dask • Airflow DAGs • MLflow • Snowflake • BigQuery • Vertex AI • Databricks • API Integration (REST/JSON)
Analytics & Visualization: Tableau • Power BI • Looker • Looker Studio • Excel • KPI Dashboards • Flourish
Tools & Platforms: GitHub • JIRA • Salesforce • GCP (Google Cloud Platform) • AWS